

Bioprospecting rice endophytic bacteria and their dual-strain consortia for plant growth promotion and leaf metabolite homeostasis in rice (*Oryza sativa* L. cv. IR-64) under graded salinity stress

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Abstract

Salinity limits rice yields on roughly one billion hectares of salt-affected land, and the seedling stage is where the crop is most exposed. We asked whether six endophytic bacteria recovered from rice (IS-01 to IS-06), and five dual-strain consortia assembled from them (C1-3, C1-5, C2-3, C2-5 and C4-5), could keep salt-stressed seedlings of the salt-susceptible cultivar IR-64 growing and metabolically stable. Seeds were bacterized and tested first on moist filter paper and then in a polyhouse pot trial at 0, 50, 100 and 150 mM NaCl. We measured germination, root and shoot length, seedling vigour index (SVI), chlorophyll *a* and *b*, carotenoids, total phenolics, soluble protein, soluble sugars and DPPH radical scavenging. Salinity cut SVI by 57.7%, chlorophyll *a* by 82.8%, soluble protein by 60.3% and soluble sugars by 67.5% at 150 mM. Inoculation did not change germination significantly in either assay (likelihood-ratio $\chi^2 = 11.61$, $df = 6$, $p = 0.071$ for isolates), but it strongly affected every continuous growth and biochemical trait (two-way ANOVA, inoculant $\times$ NaCl interaction $p < 0.001$ for all traits except consortium shoot length). At 150 mM NaCl, IS-06 raised chlorophyll *a* by 240%, soluble sugars by 169% and SVI by 142% relative to the salinity-matched uninoculated control. Consortium C4-5 (IS-04 + IS-05) kept 80.8% of its unstressed protein level, against 39.7% in uninoculated plants, and exceeded both parent strains for protein, chlorophyll *a* and SVI. Principal component analysis placed 51.7% of the variance on a single growth–metabolite axis, and a composite membership index ranked IS-06 (0.647) and C4-5 (0.617) first under severe stress. Not every pairing helped: C1-3, C1-5 and C2-3 ranked below both of their parent strains. We therefore recommend IS-06 and C4-5 for multi-location field testing, with strain identity and compatibility confirmed before formulation.

Keywords: bacterial endophytes; microbial consortium; seed bacterization; salinity tolerance; osmolytes; photosynthetic pigments; antioxidant activity; rice

Highlights

- Six rice endophytes and five dual-strain consortia were screened under 0 to 150 mM NaCl.
- Salinity reduced seedling vigour by 57.7% and leaf soluble sugars by 67.5% at 150 mM.
- Inoculation did not alter germination significantly but rescued growth and leaf metabolites.
- Isolate IS-06 and consortium C4-5 ranked highest on a composite resilience index.
- Consortium benefit depended on the partner pair; three of five pairings underperformed.

1. Introduction

Rice feeds more than half of the world's population, yet it is one of the more salt-sensitive cereals, and that sensitivity peaks during seedling establishment and again at flowering (Singh et al., 2021). The problem is not small. About one billion hectares of land worldwide carry some form of salinity or sodicity (Basak et al., 2022), and irrigated rice systems in South and Southeast Asia sit squarely inside that footprint (Saleem et al., 2025; Sackey et al., 2025). Breeding has produced tolerant cultivars, but progress is slow because tolerance is controlled by many genes acting at different stages (Ismail & Horie, 2017). Farmers on marginal saline land need something they can use this season.

Salt damages plants in two overlapping phases. The first is osmotic: a saline root zone lowers the water potential of the soil solution, so roots struggle to take up water and cell expansion slows within hours (Munns & Tester, 2008). The second phase is ionic. Sodium and chloride accumulate in older leaves, displace potassium, disturb enzyme function and speed up leaf senescence (Munns & Tester, 2008; van Zelm et al., 2020). Riding on both phases is oxidative stress. When CO₂ fixation slows, electrons pile up in the chloroplast and reactive oxygen species (ROS) form faster than the antioxidant system can remove them (Gill & Tuteja, 2010). Membranes, pigments and proteins are damaged in the process. Plants respond by adjusting signalling, ion transport and metabolism (Zhu, 2016), and by accumulating compatible solutes such as soluble sugars, proline and glycine betaine (Parida & Das, 2005; Mahajan & Tuteja, 2005). The size of these adjustments differs between genotypes, and for that reason physiological markers like chlorophyll, soluble protein and sugars are widely used to score salt responses (Negrão et al., 2017).

Endophytic bacteria offer a different route to tolerance. By definition they live inside plant tissues for at least part of their life cycle without causing disease (Hardoim et al., 2015). Rice roots, stems, leaves and seeds carry a diverse endophytic community, with *Bacillus*,

Pseudomonas, *Enterobacter* and *Pantoea* among the genera most often isolated (Santoyo et al., 2016; Moronta-Barrios et al., 2018; Afzal et al., 2019). Because they already live inside the host, endophytes are thought to face less competition from the soil microflora than rhizosphere inoculants do (Compant et al., 2019). Seed-borne endophytes are particularly interesting. They travel with the seed, colonize the seedling early and can be passed on vertically (Walitang et al., 2017). Their composition in rice shifts with host genotype and with salt stress itself (Walitang et al., 2018), which hints that some of these bacteria have been selected for life in a saline host.

How do these bacteria help? The best-studied mechanism is the enzyme 1-aminocyclopropane-1-carboxylate (ACC) deaminase. It breaks down ACC, the immediate precursor of ethylene, and so blunts the burst of stress ethylene that otherwise inhibits root elongation (Glick, 2014). Growth-promoting bacteria also produce indole-3-acetic acid, solubilize phosphate and other nutrients, release siderophores and, in some cases, fix nitrogen (Glick, 2012; Etesami & Maheshwari, 2018). Under salinity, a further set of traits becomes relevant. Inoculated rice plants have shown lower shoot Na^+ , better K^+ uptake, higher antioxidant enzyme activity and altered hormone profiles (M. A. Khan et al., 2021; Asif et al., 2023; Siddika et al., 2024). Exopolysaccharide-producing endophytes can bind sodium around the roots (Sun et al., 2020), and inoculated seedlings often accumulate more soluble sugars and other osmolytes (Arora et al., 2020; Ramasamy & Mahawar, 2023). Halotolerant *Enterobacter* strains carrying ACC deaminase have promoted rice seedling growth under salt (Sarkar et al., 2018; Sagar et al., 2020), and salt-tolerant rhizobacteria have improved both growth and grain yield of rice in saline soil (Shultana et al., 2020). Several reviews now treat microbe-mediated salt tolerance as a stack of complementary mechanisms rather than a single trait (Ilangumaran & Smith, 2017; Numan et al., 2018; Etesami & Glick, 2020; Zamanzadeh-Nasrabadi et al., 2023).

That view leads naturally to consortia. If no single strain carries every useful trait, combining two or more compatible strains might cover more of the stress response. A meta-analysis of 51 live-soil studies found that consortium inoculation raised plant growth by 48% on average, compared with 29% for single-strain inoculation (Liu et al., 2023). Consortia have also improved wheat growth and yield under saline conditions (M. Y. Khan et al., 2022), and a ten-member synthetic community colonized rice roots together as a stable group (Moronta-Barrios et al., 2018). The case is not closed, though. Strains can compete for the same niche, and poorly matched partners may cancel each other out. Most rice studies still

test either a single strain or a single mixture (Ganie et al., 2021; Wang et al., 2025). Comparisons that line up individual isolates next to consortia built from those same isolates, across germination, seedling growth and leaf biochemistry, are scarce.

A second gap is methodological. Screening papers often report dozens of traits one at a time, each with its own significance test, and rank inoculants by whichever trait looks best. That approach makes it hard to see whether an inoculant helps the plant as a whole, or only moves one number. Standardizing traits and combining them, through indices and multivariate ordination, gives a fairer picture (Negrão et al., 2017; Jolliffe & Cadima, 2016).

This study was built around those two gaps. Our objectives were to (i) assemble dual-strain consortia from six rice endophytes and check their in vitro compatibility; (ii) compare isolates and consortia for their effects on germination and seedling vigour in a laboratory assay; (iii) test the same inoculants in a polyhouse pot trial under 0, 50, 100 and 150 mM NaCl, measuring growth, photosynthetic pigments, phenolics, soluble protein, soluble sugars and DPPH radical scavenging; and (iv) rank the inoculants with salinity-induced reduction and mitigation indices, a composite membership index and principal component analysis. We expected at least one consortium to outperform both of its parent strains under severe salinity. We also expected that the benefit of mixing would depend on which strains were paired.

2. Materials and methods

2.1 Experimental overview

The work ran in five linked stages, summarized in Fig. 1. Inocula and consortia were prepared first. Rice seeds were then surface-sterilized and bacterized. A laboratory germination assay screened all inoculants without salt, and a polyhouse pot trial tested them across a graded NaCl series. Leaf tissue from the pot trial supplied every biochemical measurement, and the combined data set was analysed with univariate, index-based and multivariate methods.

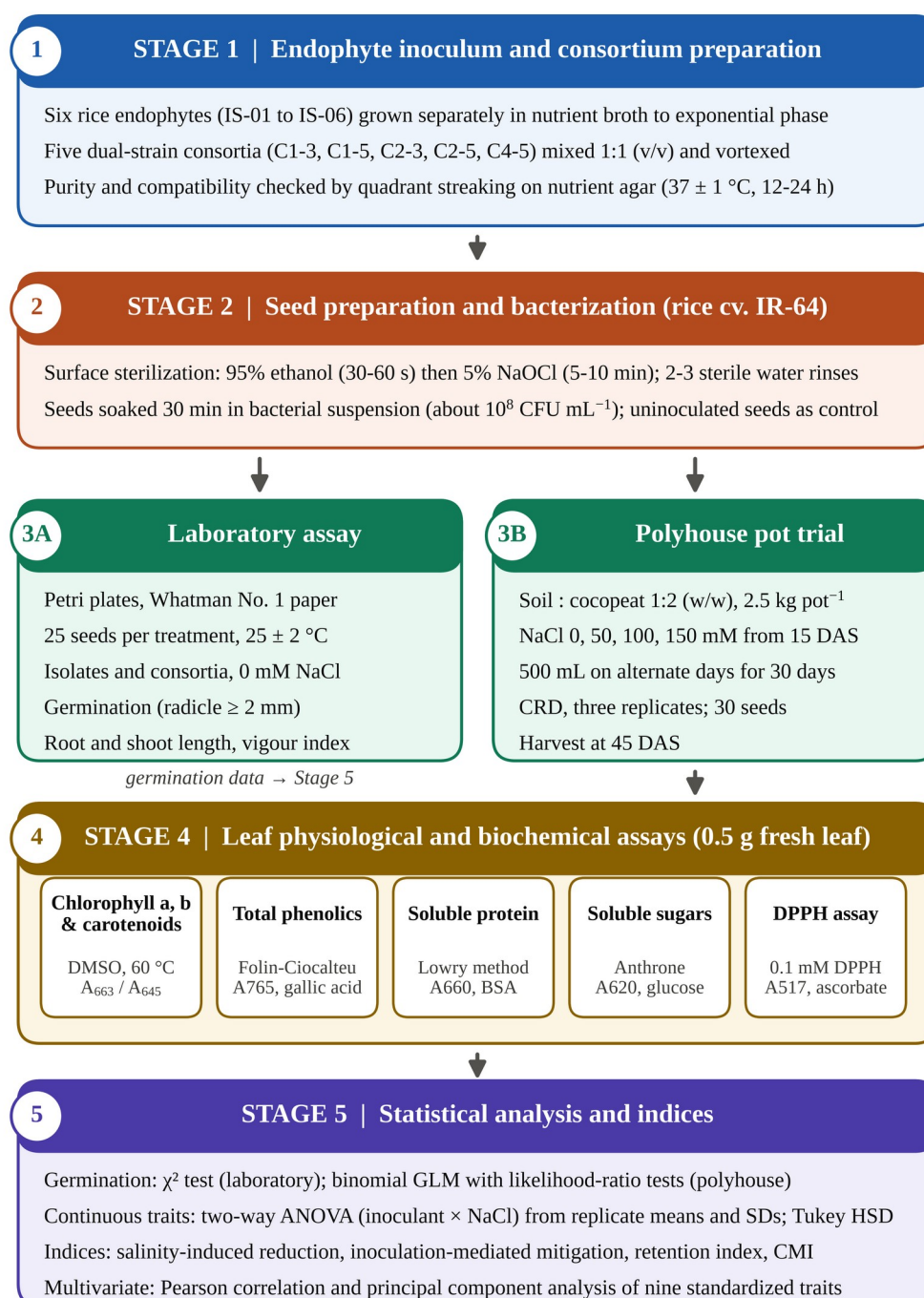


Fig. 1. Infographic flowchart of the experimental workflow. Stage 1, preparation of single-isolate inocula and dual-strain consortia with purity checks; Stage 2, surface sterilization and bacterization of rice cv. IR-64 seeds; Stage 3A, laboratory germination assay; Stage 3B, polyhouse pot trial under 0–150 mM NaCl; Stage 4, leaf biochemical assays on polyhouse plants; Stage 5, statistical analysis, derived indices and multivariate analysis. DAS, days after sowing; CRD, completely randomized design; CMI, composite membership index.

2.2 Endophytic isolates and consortium preparation

Six endophytic bacterial isolates previously recovered from surface-sterilized tissues of rice (*Oryza sativa* L.) plants, coded IS-01 to IS-06, were used. Each isolate was grown

separately in nutrient broth at 37 ± 1 °C until the culture reached exponential phase. Five two-member consortia were assembled by mixing equal volumes (1:1, v/v) of the relevant cultures under aseptic conditions: C1-3 (IS-01 + IS-03), C1-5 (IS-01 + IS-05), C2-3 (IS-02 + IS-03), C2-5 (IS-02 + IS-05) and C4-5 (IS-04 + IS-05). Each mixture was vortexed gently so that both partners were evenly distributed. To check compatibility and purity, a loopful of every consortium was streaked by the quadrant method on nutrient agar and incubated at 37 ± 1 °C for 12–24 h. Colony morphology, density and any zone of inhibition between colony types were recorded, following the general approach used for rice seed endophytes by Walitang et al. (2017).

2.3 Plant material, seed sterilization and bacterization

Seeds of IR-64, a widely grown but salt-susceptible *indica* rice cultivar, were used throughout. Uniform, healthy seeds were immersed in 95% (v/v) ethanol for 30–60 s with gentle swirling, transferred to 5% (v/v) sodium hypochlorite for 5–10 min and rinsed two to three times in sterile distilled water. Sterilized seeds were used immediately. For bacterization, seeds were soaked for 30 min in the appropriate bacterial suspension (about 10^8 CFU mL⁻¹); uninoculated control seeds were not bacterized.

2.4 Preparation of NaCl solutions

Sodium chloride (molar mass 58.44 g mol⁻¹) was dissolved in sterile distilled water to give 50 mM (2.922 g L⁻¹), 100 mM (5.844 g L⁻¹) and 150 mM (8.766 g L⁻¹) solutions. Sterile distilled water served as the 0 mM treatment.

2.5 Laboratory germination assay

Bacterized seeds were placed on sterile Whatman No. 1 filter paper in Petri dishes of uniform size, 25 seeds per treatment. Plates received equal volumes of sterile water so that moisture was comparable, and were kept at 25 ± 2 °C. Plates were checked regularly and topped up to prevent drying. A seed was counted as germinated when the radicle reached at least 2 mm. At the end of the observation period, root and shoot lengths were measured on the germinated seedlings. The isolate and consortium sets were run as separate batches, each with its own uninoculated control. Germination (G, %) was calculated as the number of germinated seeds divided by the number of seeds placed, multiplied by 100. The seedling vigour index was computed as $SVI = G \times (\text{root length} + \text{shoot length, cm})$, following Abdul-Baki and Anderson (1973).

2.6 Polyhouse pot experiment

The pot trial used a completely randomized design with three replicate pots per treatment. Sterilized soil and cocopeat were mixed at 1:2 (w/w), and each pot (15 cm × 18 cm) held 2.5 kg of this medium. Soil was sterilized beforehand to limit interference from resident microorganisms. Bacterized seeds of each treatment were sown in their pots, 30 seeds per treatment in total, and the pots were irrigated regularly. The isolate experiment comprised seven inoculation levels (uninoculated, IS-01 to IS-06) and the consortium experiment six (uninoculated, C1-3, C1-5, C2-3, C2-5, C4-5), each crossed with four NaCl levels (0, 50, 100 and 150 mM). Salt stress was applied by irrigating each pot with 500 mL of the relevant solution on alternate days for 30 days, beginning at the two- to three-leaf stage (15 days after sowing, DAS). Germination in pots was scored as the percentage of the 30 sown seeds that produced a seedling. After establishment, seedlings were thinned to three uniform plants per pot. Plants were harvested at 45 DAS, when root and shoot lengths were measured and fully expanded leaves were sampled for biochemistry. SVI was calculated as in Section 2.5.

2.7 Photosynthetic pigments

Fresh leaf tissue (0.5 g) was placed in 5 mL of dimethyl sulfoxide (DMSO) and incubated at 60 °C for 30 min, a non-maceration extraction adapted from Hiscox and Israelstam (1979). Absorbance was read at 663 and 645 nm, and at 490 nm for carotenoids, against a DMSO blank on a UV–visible spectrophotometer. Chlorophyll *a* and *b* were calculated with Arnon's (1949) equations: $\text{Chl } a \text{ (mg L}^{-1}\text{)} = 12.7 A_{663} - 2.69 A_{645}$; $\text{Chl } b \text{ (mg L}^{-1}\text{)} = 22.9 A_{645} - 4.68 A_{663}$. Values were converted to a fresh-weight basis as $C \times V / W$, where *C* is pigment concentration, *V* extract volume and *W* tissue fresh weight. Total chlorophyll was taken as the sum of chlorophyll *a* and *b*. Carotenoid determination followed the general principles set out by Lichtenthaler (1987).

2.8 Total phenolic content

Fresh leaf tissue (0.5 g) was homogenized in 80% (v/v) methanol and centrifuged, and the clear supernatant was used. Total phenolics were estimated with the Folin–Ciocalteu reagent as described by Singleton and Rossi (1965). An aliquot of extract (0.5 mL) was made up to 1.0 mL with distilled water, mixed with 0.5 mL of Folin–Ciocalteu reagent (diluted 1:1 with water) and then with 0.5 mL of 20% (w/v) sodium carbonate. After 30–60 min at room temperature, 7.0 mL of distilled water was added and absorbance was read at 765 nm. A

gallic acid calibration (20–100 $\mu\text{g mL}^{-1}$) gave $y = 0.009x + 0.005$ ($R^2 = 0.9998$). Results are expressed as μg gallic acid equivalents per mL of extract. Because every sample was prepared from the same tissue mass in the same extraction volume, these values are directly comparable between treatments.

2.9 Soluble protein

Protein was extracted from 0.5 g of fresh leaf tissue and estimated by the method of Lowry et al. (1951), using bovine serum albumin as the standard (40–200 $\mu\text{g mL}^{-1}$). An aliquot of extract (0.5 mL) was made up to 1.0 mL, mixed with 5.0 mL of freshly prepared alkaline copper reagent (2% Na_2CO_3 in 0.1 N NaOH with 2% sodium potassium tartrate, plus 1% CuSO_4 at 50:1) and left for 10 min. Diluted Folin–Ciocalteu reagent (0.5 mL) was then added rapidly, and absorbance was read at 660 nm after a further 10 min. Protein is expressed as $\mu\text{g mL}^{-1}$ of extract.

2.10 Soluble sugars

Soluble sugars were extracted from 0.5 g of fresh leaf tissue in hot 80% (v/v) ethanol and estimated with anthrone (Yemm & Willis, 1954). One millilitre of extract was mixed with 5.0 mL of freshly prepared anthrone reagent (about 0.2%, w/v, in diluted H_2SO_4), heated in a boiling water bath for 10 min, cooled quickly and read at 620 nm. A glucose calibration (20–100 $\mu\text{g mL}^{-1}$) was used, and values are expressed as μg glucose equivalents per mL of extract.

2.11 DPPH radical scavenging activity

Methanolic leaf extracts (0.5 mL, made up to 1.0 mL with methanol) were mixed with 3.0 mL of 0.1 mM 2,2-diphenyl-1-picrylhydrazyl (DPPH) in methanol, wrapped in foil and incubated in the dark for 30 min at room temperature (Brand-Williams et al., 1995). Absorbance was read at 517 nm against a DPPH control without extract, and ascorbic acid (25–400 $\mu\text{g mL}^{-1}$) served as the reference antioxidant. Scavenging activity was calculated as $[(A_0 - A_1) / A_0] \times 100$, where A_0 and A_1 are the absorbances of the control and the sample (Gülçin, 2020). The isolate and consortium extracts were assayed in separate batches.

2.12 Derived indices

Three simple indices were computed from treatment means. The salinity-induced reduction (SIR, %) described damage in uninoculated plants: $\text{SIR} = (X_0 - X_S) / X_0 \times 100$, where X_0 and X_S are trait values at 0 mM and at salinity level S. The inoculation-mediated

mitigation (IMM, %) compared each inoculant with the uninoculated control at the same salinity: $IMM = (X_{i,s} - X_{c,s}) / X_{c,s} \times 100$. The retention index (RI, %) expressed how much of its own unstressed value an inoculant kept at 150 mM: $RI = X_{i,150} / X_{i,0} \times 100$. Relative measures of this kind are preferred when genotypes or treatments differ in their unstressed baseline (Negrão et al., 2017).

To summarize performance across traits, we built a composite membership index (CMI). Each of nine traits (germination, root length, shoot length, SVI, chlorophyll *a*, carotenoids, soluble protein, soluble sugars and DPPH scavenging) was rescaled to 0–1 across all 48 treatment combinations as $(X - X_{min}) / (X_{max} - X_{min})$, and the nine rescaled values were averaged. A CMI near 1 means a treatment sat near the top of the observed range for most traits. Total phenolics and chlorophyll *b* were left out of the CMI: phenolic data were incomplete for the isolate set, and chlorophyll *b* values differed by an order of magnitude between the two experimental batches (Section 3.4).

2.13 Statistical analysis

Germination counts were analysed as binomial data. For the laboratory assay, a χ^2 test of independence compared germinated and non-germinated seeds across treatments within each batch, and Fisher's exact test compared the best inoculant with the control. For the pot trial, a binomial generalized linear model with a logit link was fitted to the germination proportions, with inoculant and NaCl level as factors. Salinity, inoculant and interaction effects were tested by likelihood-ratio (LR) χ^2 tests between nested models.

Continuous traits were analysed by two-way analysis of variance (ANOVA) with inoculant and NaCl level as fixed factors. Because the data were available as replicate means and standard deviations ($n = 3$), sums of squares were computed directly from these sufficient statistics (Larson, 1992). The error mean square was pooled from the within-treatment variances, and partial eta-squared (η_p^2) was reported as the effect size. Tukey's honestly significant difference (HSD) at $\alpha = 0.05$ was calculated for comparisons among the 28 (isolate) or 24 (consortium) treatment means (Tukey, 1949). Traits for which replicate SDs were not recorded (consortium soluble sugars and consortium DPPH activity) were described without inferential tests.

Pearson correlations were computed among the nine traits used in the CMI across the 48 treatment means; with 46 degrees of freedom, $|r| \geq 0.285$ corresponds to $p < 0.05$ and $|r| \geq 0.465$ to $p < 0.001$. Principal component analysis (PCA) was run on the standardized (z-

scored) trait matrix (Jolliffe & Cadima, 2016). All analyses were carried out in Python with SciPy (Virtanen et al., 2020) and statsmodels (Seabold & Perktold, 2010). Unless stated otherwise, values are means $\pm$ SD of three replicates.

3. Results

3.1 Consortium compatibility

All five pairings grew as uniform, stable cultures on nutrient agar. We saw no inhibition zones, no pigment changes and no odd colony forms where partner strains met, so there was no visible sign of antagonism. Colony density on the consortium plates looked higher than on plates of the single isolates, and growth appeared faster. These are qualitative observations. They show that the partners can share a plate; they don't show that the partners cooperate inside the plant.

3.2 Laboratory germination and early seedling vigour

Germination on filter paper varied from 60% to 92% among the single isolates and from 56% to 84% among the consortia (Table 1). IS-06 gave the highest germination (92%) and the longest shoots (92.00 ± 2.64 mm). IS-04 gave the longest roots of the isolate set (73.00 ± 4.35 mm). Once root and shoot length were combined into SVI, IS-06 (1327.8) and IS-04 (1313.2) stood well above the uninoculated control (920.2). IS-02 and IS-03 fell below it (545.9 and 507.7). Among the consortia, C4-5 produced the longest roots of any treatment in the assay (90.66 ± 0.57 mm) and the highest consortium SVI (1271.9), with C2-5 close behind (1215.1). C1-3 and C1-5 germinated less well than their own batch control.

The germination differences themselves were not statistically significant. Proportions did not differ across the isolate treatments ($\chi^2 = 9.40$, $df = 6$, $p = 0.152$) or across the consortium treatments ($\chi^2 = 7.02$, $df = 5$, $p = 0.219$). Even the largest single contrasts, IS-06 against its control (92% vs. 80%) and C2-5 against its control (84% vs. 68%), were not significant by Fisher's exact test ($p = 0.417$ and $p = 0.321$). With 25 seeds per treatment, the assay could only have detected much larger differences. The strong effects in this screen were on seedling growth, not on whether seeds germinated.

Table 1. Germination and early seedling growth of rice cv. IR-64 bacterized with single endophytic isolates or dual-strain consortia in the laboratory assay (0 mM NaCl; 25 seeds per treatment).

Treatment	Germinated / total	Germination (%)	Root length (mm)	Shoot length (mm)	SVI
<i>Single isolates (batch 1)</i>					
Control	20 / 25	80	62.43 ± 1.19	52.60 ± 3.78	920.2
IS-01	19 / 25	76	55.66 ± 3.05	34.00 ± 4.58	681.4
IS-02	15 / 25	60	40.33 ± 0.47	50.66 ± 1.15	545.9
IS-03	17 / 25	68	32.33 ± 2.08	42.33 ± 1.52	507.7
IS-04	21 / 25	84	73.00 ± 4.35	83.33 ± 3.21	1313.2
IS-05	20 / 25	80	63.00 ± 1.00	72.33 ± 2.51	1082.6
IS-06	23 / 25	92	52.33 ± 2.51	92.00 ± 2.64	1327.8
<i>Dual-strain consortia (batch 2)</i>					
Control	17 / 25	68	43.50 ± 1.09	33.78 ± 2.30	525.5
C1-3	14 / 25	56	70.00 ± 0.00	44.00 ± 3.60	638.4
C1-5	15 / 25	60	63.66 ± 1.52	51.66 ± 2.88	691.9
C2-3	17 / 25	68	41.66 ± 2.08	33.33 ± 3.21	509.9
C2-5	21 / 25	84	83.66 ± 2.30	61.00 ± 1.73	1215.1
C4-5	20 / 25	80	90.66 ± 0.57	68.33 ± 2.08	1271.9

Root and shoot lengths are means ± SD. SVI = germination (%) × (root + shoot length, cm), recalculated from the recorded values (Abdul-Baki & Anderson, 1973). Germination did not differ among treatments within either batch (isolates: $\chi^2 = 9.40$, df = 6, $p = 0.152$; consortia: $\chi^2 = 7.02$, df = 5, $p = 0.219$).

3.3 Germination and seedling growth under salinity in the polyhouse

Salinity lowered germination in the pots in a clear, dose-dependent way (Fig. 2; Table 2). Uninoculated seeds fell from 83.3% at 0 mM to 60.0% at 150 mM, an SIR of 28.0%. The binomial model confirmed a strong salinity effect in both experiments (isolates: LR $\chi^2 = 34.84$, df = 3, $p < 0.001$; consortia: LR $\chi^2 = 23.15$, df = 3, $p < 0.001$). Inoculation shifted germination upward in most treatments. IS-06 germinated at 93.3%, 90.0%, 76.7% and 70.0% across the four salinity levels, and C4-5 at 90.0%, 93.3%, 80.0% and 76.7%. These shifts did not reach significance (isolates: LR $\chi^2 = 11.61$, df = 6, $p = 0.071$; consortia: LR $\chi^2 = 10.40$, df = 5, $p = 0.065$), and there was no evidence of an inoculant × salinity interaction on germination ($p = 0.999$ and $p = 0.835$).

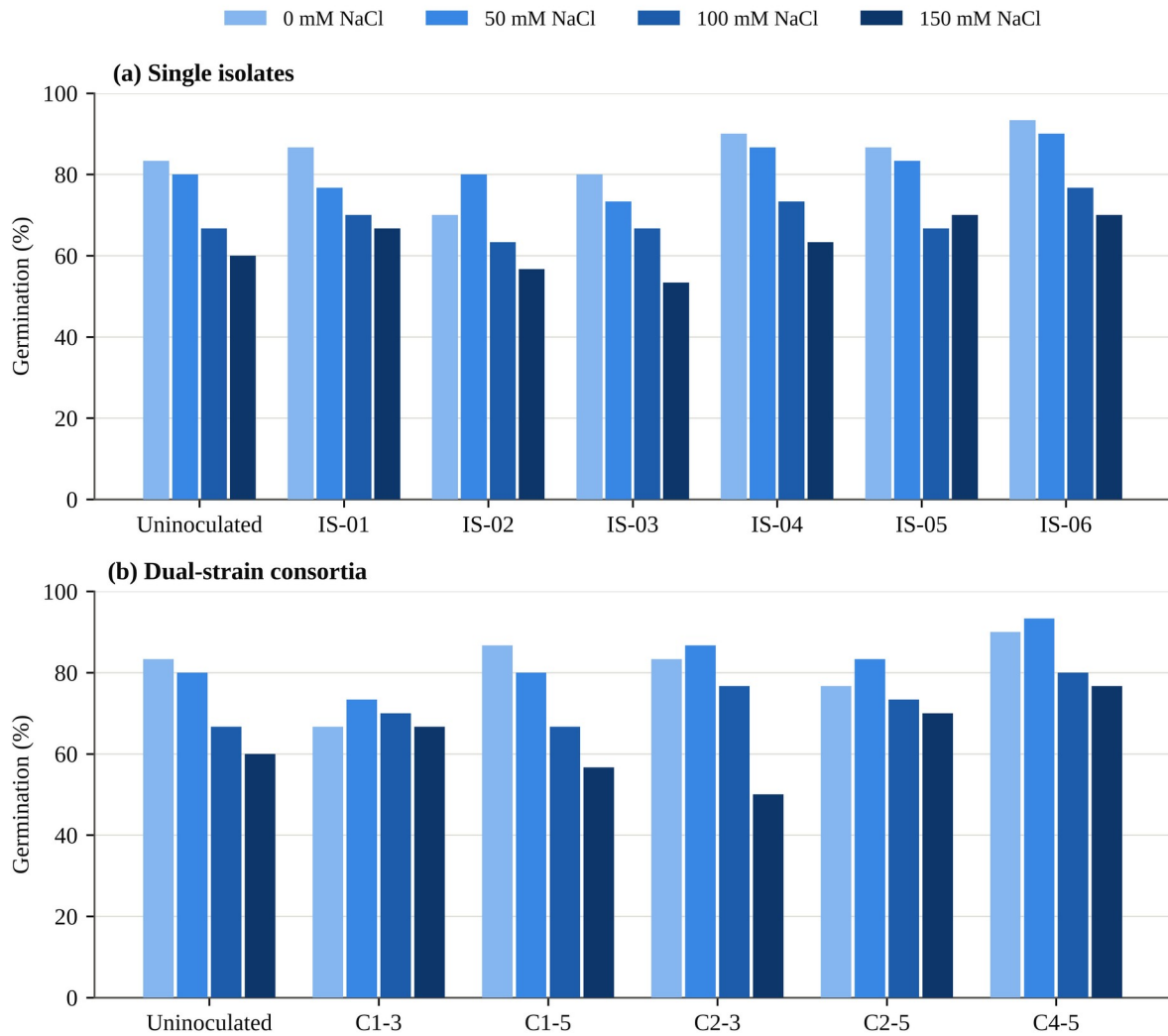


Fig. 2. Germination (%) of rice cv. IR-64 under 0, 50, 100 and 150 mM NaCl in the polyhouse after bacterization with (a) single endophytic isolates and (b) dual-strain consortia. Each bar is the percentage of 30 sown seeds. Salinity reduced germination ($p < 0.001$ in both experiments), whereas inoculant effects were not significant ($p = 0.071$ and 0.065 ; binomial GLM likelihood-ratio tests).

Seedling growth told a much stronger story. Root and shoot length of uninoculated plants dropped from 4.30 and 22.96 cm at 0 mM to 2.06 and 13.96 cm at 150 mM (SIR 52.1% and 39.2%). SVI, which combines germination with growth, fell by 57.7%. In the isolate experiment, the two-way ANOVA found large effects of inoculant, salinity and their interaction on root length (interaction $F_{18,56} = 706.4$, $p < 0.001$, $\eta_p^2 = 0.996$; HSD = 0.21 cm) and shoot length (interaction $F_{18,56} = 414.6$, $p < 0.001$; HSD = 0.68 cm) (Table 3). At 150 mM, IS-06 seedlings had roots of 5.02 ± 0.02 cm and shoots of 28.19 ± 0.26 cm, which is 144% and 102% longer than the matched control. IS-05 matched IS-06 for root length (5.03 ± 0.03 cm) but had shorter shoots (23.68 ± 0.59 cm). One value stands out in the IS-05 series: root length at 100 mM was only 1.08 ± 0.08 cm, far below both the 50 mM and the 150 mM

316 values (Fig. 3d). We report it as recorded, but it looks like an outlier rather than a biological
317 response.

318 Consortium effects on root length were also significant (interaction $F_{15,48} = 52.2$, $p <$
319 0.001 ; HSD = 0.59 cm). C4-5 gave the longest roots under stress, 5.44 ± 0.12 cm at 50 mM
320 and 4.30 ± 0.15 cm at 150 mM. Shoot length in the consortium experiment was a different
321 matter. Several salt treatments carried replicate SDs of 8 to 13 cm around means of 16 to 25
322 cm (for example, 25.0 ± 13.0 cm for C4-5 at 50 mM), and with that much spread the
323 ANOVA detected no effect of inoculant, salinity or their interaction (all $p \geq 0.39$). Unstressed
324 consortium plants were also shorter than the uninoculated control, which lowered their SVI at
325 0 mM. Under salt, though, C4-5 kept SVI between 2235 and 2841 at every stress level. That
326 is 135% above the control at 150 mM, close to the 142% achieved by IS-06 (Table 2).

327 **Table 2.** Germination (%) and seedling vigour index (SVI) of rice cv. IR-64 bacterized with endophytic isolates or consortia and grown under
328 graded NaCl in the polyhouse (30 seeds per treatment).

Treatment	Germination (%)				Seedling vigour index			
	0 mM	50 mM	100 mM	150 mM	0 mM	50 mM	100 mM	150 mM
Uninoculated control	83.3	80.0	66.7	60.0	2272	2008	1507	961
<i>Single isolates</i>								
IS-01	86.7	76.7	70.0	66.7	2724	2233	1436	1405
IS-02	70.0	80.0	63.3	56.7	2023	2201	1273	1080
IS-03	80.0	73.3	66.7	53.3	1959	1963	1765	1412
IS-04	90.0	86.7	73.3	63.3	2543	2442	1713	1766
IS-05	86.7	83.3	66.7	70.0	2895	2682	1541	2010
IS-06	93.3	90.0	76.7	70.0	2941	2751	2111	2325
<i>Dual-strain consortia</i>								
C1-3	66.7	73.3	70.0	66.7	1409	1871	1626	1555
C1-5	86.7	80.0	66.7	56.7	1629	2114	1671	1424
C2-3	83.3	86.7	76.7	50.0	1402	2169	1459	986
C2-5	76.7	83.3	73.3	70.0	1444	1925	1958	1753
C4-5	90.0	93.3	80.0	76.7	1947	2841	2235	2246

329 Germination = germinated seeds / 30 × 100. SVI = germination (%) × (root + shoot length, cm), recalculated from the recorded root and shoot
330 lengths. Binomial GLM likelihood-ratio tests, isolate experiment: NaCl $\chi^2 = 34.84$ (df 3, $p < 0.001$), inoculant $\chi^2 = 11.61$ (df 6, $p = 0.071$),
331 interaction $\chi^2 = 5.14$ (df 18, $p = 0.999$); consortium experiment: NaCl $\chi^2 = 23.15$ (df 3, $p < 0.001$), inoculant $\chi^2 = 10.40$ (df 5, $p = 0.065$),
332 interaction $\chi^2 = 9.75$ (df 15, $p = 0.835$). The uninoculated control was shared by both experiments.

3.4 Photosynthetic pigments

Chlorophyll was the trait hit hardest by salt. In uninoculated plants, chlorophyll *a* fell from 32.0 ± 0.03 to 10.0 ± 0.10 mg g⁻¹ FW at 50 mM, and to 5.5 ± 0.33 mg g⁻¹ FW at 150 mM (SIR 68.8% and 82.8%). Even the mildest salt level removed more than two-thirds of the pigment. Total chlorophyll (*a* + *b*) fell from 67.5 to 7.9 mg g⁻¹ FW. Most of the damage had already occurred at 50 mM; the curve flattened afterwards (Fig. 3a).

Inoculation held chlorophyll up at every salt level (isolate interaction $F_{18,56} = 792.0$, $p < 0.001$; HSD = 0.81 mg g⁻¹ FW). IS-06 was the most effective isolate by a wide margin. Its chlorophyll *a* values were 28.8, 25.2 and 18.7 mg g⁻¹ FW at 50, 100 and 150 mM, which is 188%, 186% and 240% above the matched control. IS-05, IS-04 and IS-01 followed, each roughly doubling chlorophyll *a* at 150 mM. Among the consortia, C4-5 (15.6 ± 0.14 mg g⁻¹ FW at 150 mM, IMM 184%) and C2-5 (12.6 ± 0.31 , IMM 129%) protected chlorophyll *a*. C1-3, C1-5 and C2-3 did not; they fell below the salinity-matched control at every salt level (Fig. 4).

Carotenoids followed the same broad pattern. Uninoculated plants dropped from 0.140 to between 0.018 and 0.032 mg g⁻¹ FW under salt. IS-06 had the highest unstressed carotenoid content (0.163 mg g⁻¹ FW) and kept 0.041 mg g⁻¹ FW at 150 mM, nearly double the control. C4-5 did better still at 150 mM (0.056 mg g⁻¹ FW, IMM 167%).

A note of caution applies to chlorophyll *b*. In the isolate experiment, chlorophyll *b* ranged from 2.4 to 35.5 mg g⁻¹ FW. In the consortium experiment, every inoculated treatment fell between 0.01 and 3.99 mg g⁻¹ FW, even at 0 mM, while the shared uninoculated control stood at 35.5 mg g⁻¹ FW. A tenfold gap of this kind is more likely to come from extraction or reading differences between batches than from biology. For that reason, we compare chlorophyll *b* only within a batch and leave it out of the indices and the multivariate analysis. Within each batch, chlorophyll *b* responded to inoculant, salinity and their interaction (isolates: interaction $F_{18,56} = 1196.0$; consortia: $F_{15,48} = 1294.0$; both $p < 0.001$).

3.5 Total phenolic content

Phenolic content of uninoculated leaves fell with salt, from 20.56 to 9.50, 7.27 and 6.15 µg mL⁻¹ at 50, 100 and 150 mM (SIR 53.8%, 64.6% and 70.1%). The complete consortium data set shows that inoculation reversed much of this loss (interaction $F_{15,48} = 430.1$, $p < 0.001$; HSD = 0.47 µg mL⁻¹). C4-5 gave the highest values at every salinity level: 26.26,

24.03, 20.67 and 16.21 $\mu\text{g mL}^{-1}$, which is 153%, 184% and 164% above the matched control under salt. C2-5 ranked second. C1-3 barely helped, and at 150 mM its value was identical to the control (6.15 $\mu\text{g mL}^{-1}$).

For the single isolates, only part of the phenolic data set could be recovered, so these values are descriptive. IS-06 contained 25.26 $\mu\text{g mL}^{-1}$ without salt and kept 24.03, 24.03 and 20.67 $\mu\text{g mL}^{-1}$ at 50, 100 and 150 mM, which is 3.4 times the control at the highest salt level. IS-05 recorded 21.79, 16.21 and 15.09 $\mu\text{g mL}^{-1}$ under the same three salt levels. The lowest value in the whole study was IS-01 at 150 mM (2.80 $\mu\text{g mL}^{-1}$), less than half the uninoculated plants at that salinity. So not every endophyte supports phenolic synthesis under salt.

3.6 Soluble protein and soluble sugars

Soluble protein declined steadily as salt rose, from 215.15 $\mu\text{g mL}^{-1}$ in unstressed controls to 133.45, 104.62 and 85.39 $\mu\text{g mL}^{-1}$ (Fig. 3b). All inoculants slowed this decline. In the isolate experiment, the ranking was the same at every salinity level: IS-06 > IS-05 > IS-04 > IS-03 > IS-02 > IS-01. At 150 mM, IS-06 held 143.06 $\mu\text{g mL}^{-1}$ (IMM 68%). The consortia did better. C4-5 recorded 215.15, 200.74 and 181.51 $\mu\text{g mL}^{-1}$ at 50, 100 and 150 mM. At 150 mM this is 113% above the control, and it is the highest protein level of any stressed treatment in the study. Expressed as the retention index, C4-5 kept 80.8% of its unstressed protein at 150 mM, C2-5 75.4% and IS-06 69.6%; uninoculated plants kept 39.7%. Inoculant, salinity and their interaction were all significant, and the effects were very large (Table 3).

Soluble sugars fell even further than protein in uninoculated plants, down to 31.83 $\mu\text{g mL}^{-1}$ at 150 mM (SIR 67.5%). Here the single isolates were the stronger group (Fig. 3c). IS-06 barely lost sugar: 96.16, 89.21, 87.47 and 85.73 $\mu\text{g mL}^{-1}$ across the four levels, a retention index of 89.2%. At 150 mM, that is 169% above the control. IS-05 and IS-04 were close behind (82.25 and 78.77 $\mu\text{g mL}^{-1}$ at 150 mM). Among the consortia, C4-5 again ranked first (99.64, 82.25 and 70.08 $\mu\text{g mL}^{-1}$ at 50, 100 and 150 mM; IMM 120% at 150 mM), but consortium sugar levels dropped more steeply with salt than those of the best isolates. Protein and sugar moved together across treatments ($r = 0.89$, $p < 0.001$).

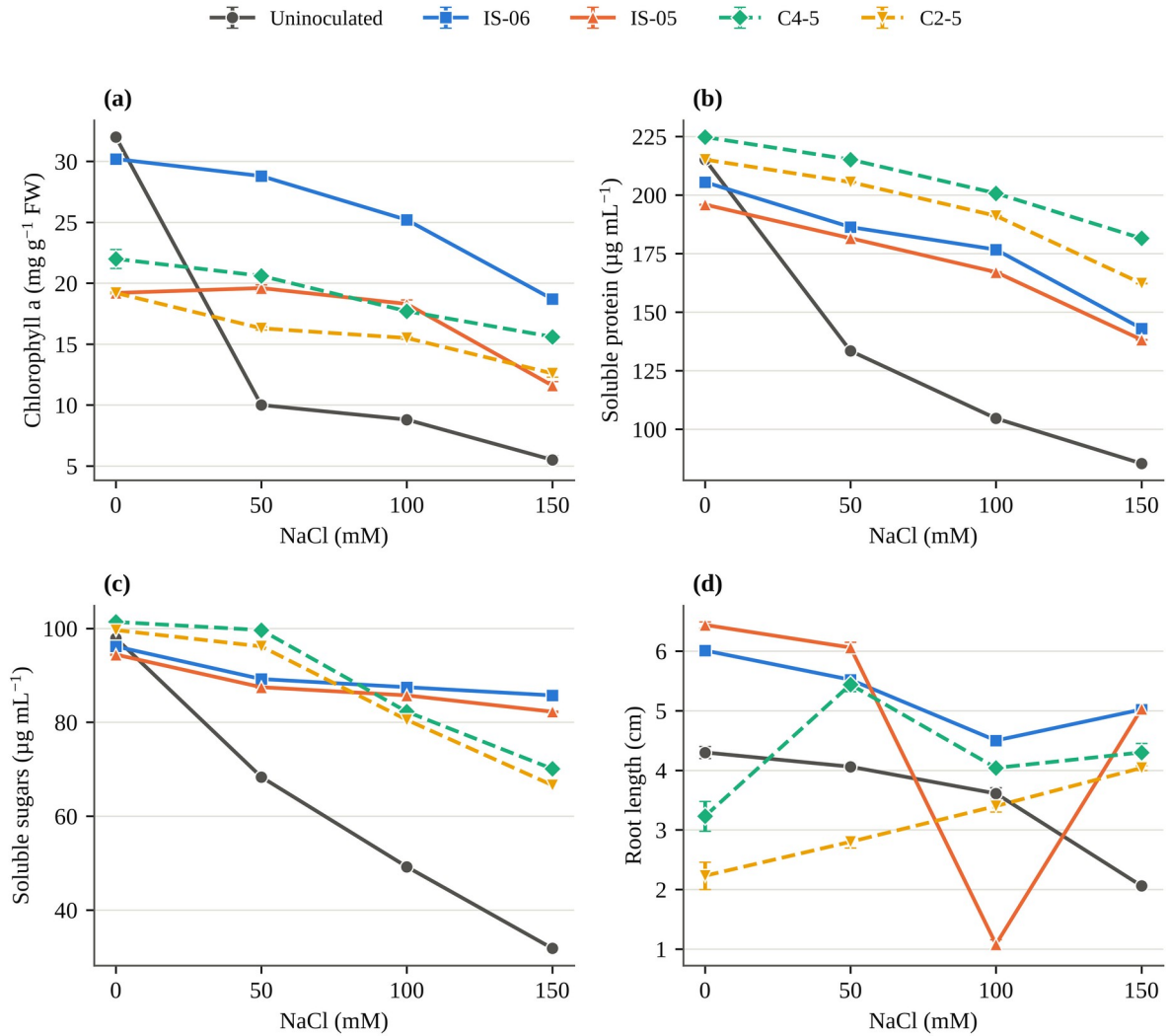


Fig. 3. Dose–response of (a) chlorophyll *a*, (b) soluble protein, (c) soluble sugars and (d) root length to NaCl in uninoculated rice and in plants inoculated with the two best single isolates (IS-06, IS-05; solid lines) and the two best consortia (C4-5, C2-5; dashed lines). Error bars are SD (n = 3); replicate SDs were not recorded for consortium sugars. The low IS-05 root value at 100 mM is shown as recorded.

3.7 DPPH radical scavenging activity

DPPH scavenging did not behave like the other traits. In the isolate batch, uninoculated leaves scavenged 90% of the radical at 0 mM, 65% at 50 mM, 82% at 100 mM and 70% at 150 mM. This is not monotonic, and the high value at 100 mM is hard to explain biologically. Without salt, every isolate gave lower activity than the control (64–80%). Under salt, the stronger isolates rose above it: IS-06 recorded 81%, 78% and 88% at 50, 100 and 150 mM, which is 26% above the control at 150 mM. IS-05 reached 80% at 150 mM. The interaction was significant ($F_{18,56} = 76.7$, $p < 0.001$; HSD = 3.19 percentage points).

In the consortium batch, the unstressed control scavenged only 70.6%, so the two batches can't be merged at 0 mM. Replicate SDs were not recorded for this batch. The pattern still matched the isolate results: C4-5 gave the highest activity at 50 mM (74.8%) and 150 mM (84.6%). C1-3 fell to 36.1% at 100 mM, the lowest value in the study. Across all 48 treatments, DPPH activity was essentially unrelated to protein ($r = -0.04$) and sugar ($r = -0.02$). It was only moderately linked to chlorophyll *a* ($r = 0.42$, $p < 0.01$).

3.8 Mitigation profile and composite ranking

Fig. 4 puts every inoculant, trait and salinity level on one heat map, expressed as IMM relative to the matched control. Three patterns stand out. First, protein and sugar were raised by every inoculant at every salinity level; not one of the 66 isolate or consortium cells for these two traits is negative. Second, pigment protection split the consortia into two groups. C4-5 and C2-5 raised chlorophyll *a* and carotenoids, while C1-3, C1-5 and C2-3 lowered them. Third, the benefits grew with stress. For IS-06, the IMM for chlorophyll *a* rose from 188% at 50 mM to 240% at 150 mM, and for soluble sugars from 31% to 169%.

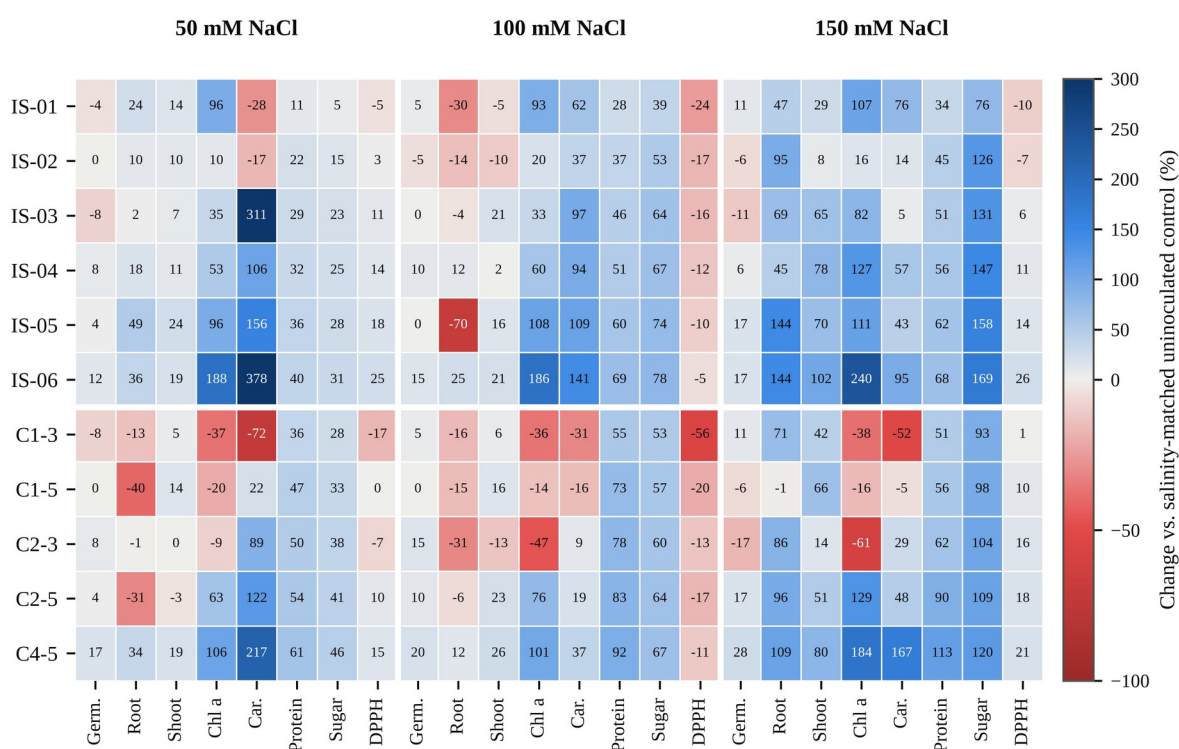


Fig. 4. Heat map of inoculation-mediated mitigation (IMM, %), i.e. the change in each trait relative to the uninoculated control at the same NaCl level, for six single isolates (upper block) and five consortia (lower block) at 50, 100 and 150 mM NaCl. Blue cells denote improvement and red cells impairment; the colour scale is truncated at -100% and +300%, and cell labels give exact values. Germ., germination; Car., carotenoids.

The CMI condenses these nine traits into one number per treatment. At 150 mM, the ranking was IS-06 (0.647), C4-5 (0.617), IS-05 (0.534), C2-5 (0.482), IS-04 (0.462) and IS-03 (0.375), followed by C1-3 (0.334), C1-5 (0.326), IS-01 (0.314), C2-3 (0.277) and IS-02 (0.276). The uninoculated control came last (0.139). Averaged over all four salinity levels, the first five positions stayed the same: IS-06 (0.757), C4-5 (0.672), IS-05 (0.625), IS-04 (0.571) and C2-5 (0.531). Two consortia, C2-3 (0.427) and C1-3 (0.384), ranked below the uninoculated control (0.439) on this average. The radar plot in Fig. 5 shows why IS-06 and C4-5 lead. Under 150 mM NaCl, both recover most of the trait profile of unstressed plants, although in different ways. IS-06 scores higher on root and shoot length, chlorophyll *a*, soluble sugars and DPPH activity, whereas C4-5 scores higher on germination, carotenoids and soluble protein.

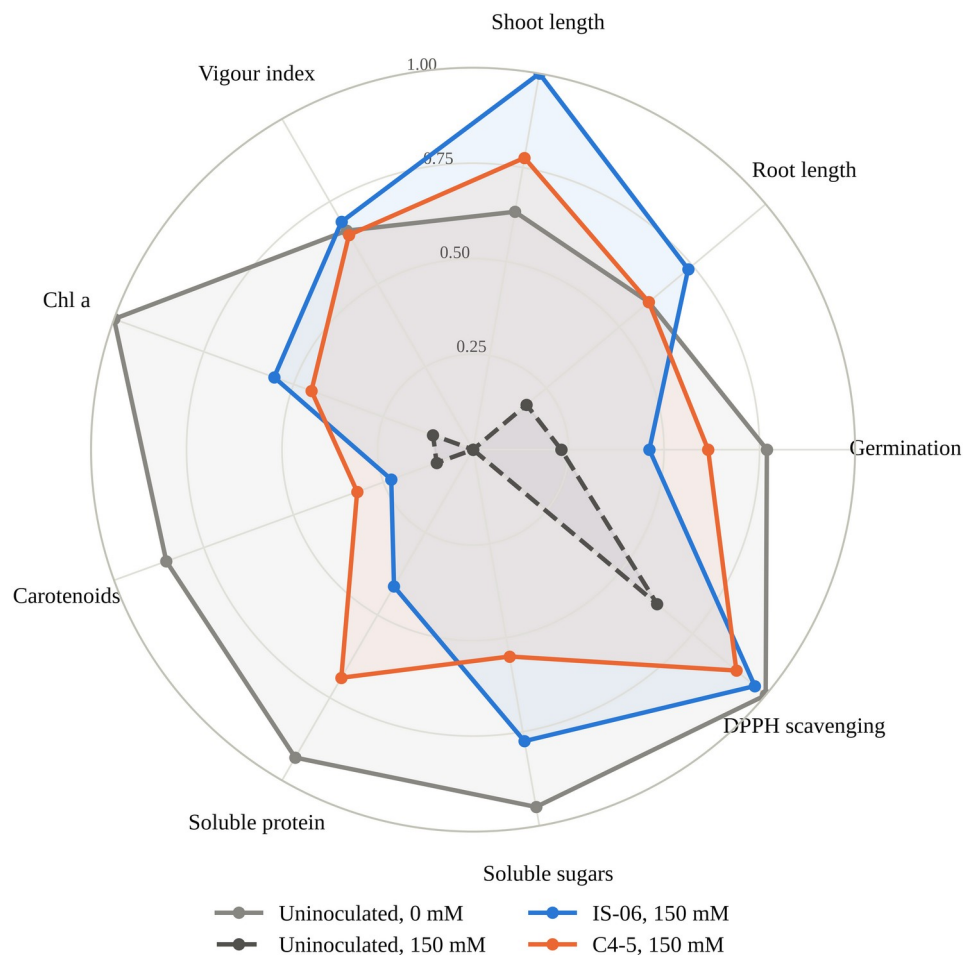
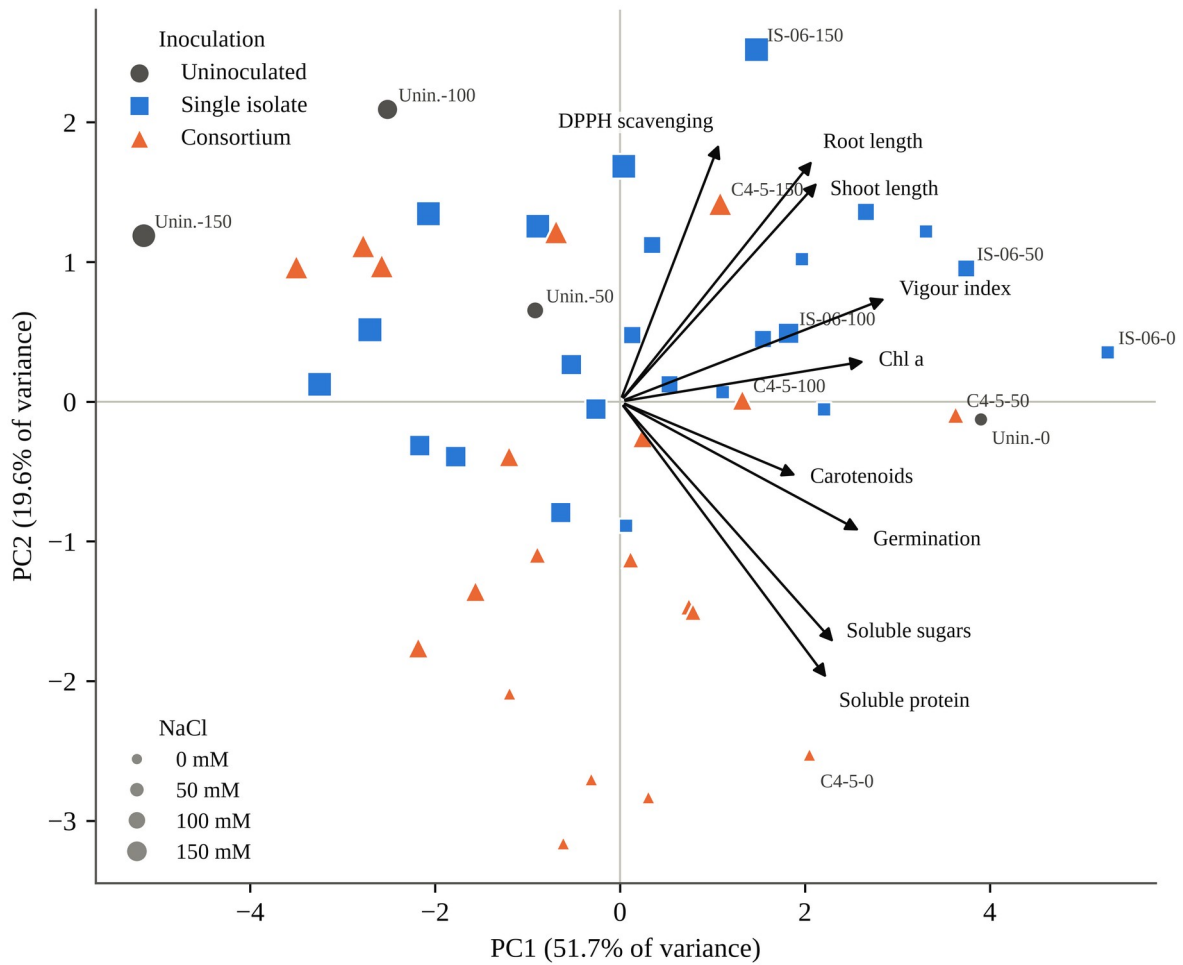


Fig. 5. Radar chart of nine traits rescaled to 0–1 across all 48 treatment combinations, comparing unstressed and 150 mM NaCl uninoculated plants with plants inoculated with IS-06 and C4-5 under 150 mM NaCl. A larger polygon indicates a trait profile closer to the best observed values.

3.9 Correlation and principal component analysis

The nine traits were largely positively correlated across treatments. SVI was tied most closely to germination ($r = 0.79$), shoot length ($r = 0.82$) and root length ($r = 0.78$). Chlorophyll *a* correlated with carotenoids ($r = 0.69$) and with SVI ($r = 0.69$). Germination tracked protein ($r = 0.73$) and sugar ($r = 0.66$) more closely than it tracked root length ($r = 0.43$).

The first two principal components explained 71.3% of the total variance (Fig. 6). PC1 (eigenvalue 4.65; 51.7%) carried positive loadings for every trait, highest for SVI (0.91), chlorophyll *a* (0.84) and germination (0.82). It is best read as a general growth-and-metabolite axis. PC2 (eigenvalue 1.77; 19.6%) separated the osmolyte traits, protein (−0.63) and sugars (−0.55), from DPPH activity (0.59) and root and shoot length (0.55 and 0.50). A third component (eigenvalue 1.13; 12.5%) also exceeded one but was not interpreted. On the biplot, uninoculated plants moved steadily towards negative PC1 as salinity rose, reaching the far left at 150 mM (PC1 = −5.1). IS-06 treatments stayed on the positive side of PC1 at every salinity level. Consortium treatments were spread widely along PC2, and the unstressed and 50 mM consortium treatments sat at its protein–sugar (negative) end, which fits their strength in protein retention. IS-06 at 150 mM sat higher on PC1 than the uninoculated control at 50 mM.



459

460 **Fig. 6.** Principal component analysis biplot of nine standardized traits across 48 treatment
 461 combinations. Symbol shape and colour indicate inoculation type and symbol size indicates
 462 NaCl level; selected treatments are labelled as inoculant–NaCl (mM). Arrows show trait
 463 loadings (scaled ×3.2). PC1 represents general seedling growth and metabolite status; PC2
 464 separates protein and sugar content from root and shoot length and DPPH activity.

465 **Table 3.** Two-way ANOVA (inoculant $\times$ NaCl) for seedling growth and leaf biochemical traits of rice cv. IR-64, computed from replicate means
466 and standard deviations (n = 3).

Trait	Inoculant F	NaCl F	Inoculant $\times$ NaCl F	η_p^2 (interaction)	Tukey HSD _{0.05}
<i>Single-isolate experiment (7 inoculation levels $\times$ 4 NaCl levels)</i>					
Root length (cm)	1156.4***	3650.8***	706.4***	0.996	0.21
Shoot length (cm)	1414.8***	1884.3***	414.6***	0.993	0.68
Chlorophyll <i>a</i> (mg g ⁻¹ FW)	4768.0***	5411.7***	792.0***	0.996	0.81
Chlorophyll <i>b</i> (mg g ⁻¹ FW)	4324.4***	5914.4***	1196.0***	0.997	0.82
Soluble protein (μ g mL ⁻¹)	1.36×10^5 ***	7.25×10^5 ***	2.75×10^4 ***	1.000	0.48
Soluble sugars (μ g mL ⁻¹)	5.61×10^4 ***	8.28×10^4 ***	9767.6***	1.000	0.48
DPPH scavenging (%)	512.5***	47.5***	76.7***	0.961	3.19
<i>Consortium experiment (6 inoculation levels $\times$ 4 NaCl levels)</i>					
Root length (cm)	123.0***	86.1***	52.2***	0.942	0.59
Shoot length (cm)	0.9 ns	1.0 ns	0.4 ns	0.106	23.21
Chlorophyll <i>a</i> (mg g ⁻¹ FW)	5916.3***	4217.1***	855.6***	0.996	0.80
Chlorophyll <i>b</i> (mg g ⁻¹ FW)	3815.2***	1753.8***	1294.0***	0.998	0.90
Soluble protein (μ g mL ⁻¹)	3.21×10^5 ***	7.72×10^5 ***	3.63×10^4 ***	1.000	0.47
Total phenolics (μ g mL ⁻¹)	1.07×10^4 ***	1.10×10^4 ***	430.1***	0.993	0.47

467 *** $p < 0.001$; ns, not significant ($p \geq 0.05$). η_p^2 , partial eta-squared. HSD is the minimum significant difference between any two treatment
468 means of the same experiment, in the units of the trait. Degrees of freedom: isolates F_{6,56} (inoculant), F_{3,56} (NaCl), F_{18,56} (interaction); consortia
469 F_{5,48}, F_{3,48}, F_{15,48}. Isolate phenolics, consortium sugars and consortium DPPH are omitted because replicate SDs were not recorded. Several traits
470 carried identical replicate SDs (± 0.15) across treatments, which inflates F; effect sizes and treatment rankings are the more reliable guide.

4. Discussion

4.1 What salinity did to IR-64 seedlings

The uninoculated plants behaved much as the literature predicts for a salt-susceptible *indica* cultivar. Growth slowed first and most visibly in the roots. At 150 mM, root length fell by about half and SVI by 57.7%, which fits the osmotic phase described by Munns and Tester (2008): water uptake becomes harder, cell expansion slows, and growth drops long before ions reach toxic levels in the leaves. The pigment data point to damage beyond osmotic slowdown. Chlorophyll *a* lost more than two-thirds of its value at only 50 mM, whereas germination at that level barely changed (80.0% vs. 83.3%). Pigment loss that steep and that early usually signals oxidative injury in the chloroplast (Gill & Tuteja, 2010). The simultaneous fall in carotenoids, which normally quench excess excitation energy, would only have made the chloroplast more vulnerable.

The metabolite results are a little less conventional. Salt-tolerant plants typically accumulate soluble sugars as osmolytes under stress (Parida & Das, 2005; Mahajan & Tuteja, 2005). In IR-64, soluble sugars moved the other way, falling by 67.5% at 150 mM, and protein and phenolics fell with them. We read this as a sign that the susceptible cultivar could not keep carbon assimilation going well enough to fund an osmotic response. With chlorophyll *a* down by 83%, less assimilate was available for sugars, amino acids and phenylpropanoid products. This interpretation is consistent with reviews that place photosynthetic capacity upstream of most other salt-tolerance traits in rice (Saleem et al., 2025; Sackey et al., 2025), but our data cannot prove the causal order. Leaf Na⁺ and K⁺, gas exchange and proline were not measured, and each would help separate these explanations.

4.2 Germination was not where inoculation made its difference

One of the clearest outcomes of this study is a negative one. Across 25 seeds on filter paper and 30 seeds per pot, no inoculant changed germination by a statistically detectable amount. Many seed bacterization studies report higher germination after inoculation (Sagar et al., 2020; Ali et al., 2022), and our direction of effect agrees with them. IS-06 and C4-5 germinated 10 to 17 percentage points above the control at 150 mM. But with 30 seeds per treatment, Fisher's exact test needs a difference of about 27 percentage points (for example, 60% against 87%) to reach significance, so our design was never going to confirm effects of

501 the size we saw. We'd rather say that plainly than lean on percentages that could easily shrink
502 with more seeds.

503 The distinction matters for how bacterization is sold to farmers. Salt at 150 mM cut
504 germination by 28%, root length by 52% and chlorophyll *a* by 83%. Germination was
505 therefore one of the least salt-sensitive traits we measured, and the stand losses that matter in
506 saline paddies are more likely to come from weak seedlings than from seeds that never
507 sprout. That is where the inoculants worked. Singh et al. (2021) point out that rice is most
508 vulnerable during seedling establishment, not at imbibition, and our data fit that pattern.
509 There is also a caveat on timing. Salt was applied from the two- to three-leaf stage onwards,
510 so salinity effects on emergence should have been small; the decline we recorded suggests
511 that early saline irrigation reached the seed zone sooner than planned, or that the counts partly
512 reflect seedling survival. Future trials should record emergence and survival separately.

513 **4.3 IS-06: a strong single strain for pigment and osmolyte protection**

514 IS-06 was the most consistent performer. Under salt, it ranked first among the isolates
515 for chlorophyll *a*, soluble protein, soluble sugars and DPPH activity at every NaCl level, first
516 or joint first for germination, and it topped the CMI at all four salinity levels. The size of its
517 effect on chlorophyll *a*, a 240% gain over the control at 150 mM, is on the high side of what
518 has been reported for rice. Halotolerant *Bacillus* strains, for instance, restored chlorophyll and
519 photosynthetic parameters of salt-stressed rice alongside better ion homeostasis and
520 antioxidant defence (Siddika et al., 2024). Halotolerant endophytes have done the same while
521 adjusting hormone levels and the expression of ion transporter genes (Asif et al., 2023).

522 The mechanism behind IS-06 is still unknown, because the isolates were not
523 characterized for growth-promoting traits here. Two routes seem most likely. The first is
524 ACC deaminase. By lowering stress ethylene, it would protect root elongation and delay
525 ethylene-driven leaf senescence and chlorophyll breakdown (Glick, 2014); rice seedlings
526 inoculated with ACC deaminase-positive *Enterobacter* strains showed exactly this kind of
527 growth protection (Sarkar et al., 2018; Sagar et al., 2020). The second route is better ionic
528 control. If IS-06 reduced sodium entry into the shoot, as reported for other rice endophytes
529 (M. A. Khan et al., 2021; Asif et al., 2023), the chloroplast would face less ionic stress and
530 pigment synthesis could continue. The two routes aren't mutually exclusive, and tests for
531 ACC deaminase activity, indole-3-acetic acid production, exopolysaccharide synthesis and
532 shoot Na^+/K^+ ratios would tell them apart.

The soluble sugar data add something. IS-06 plants kept 89% of their unstressed sugar level at 150 mM, whereas uninoculated plants kept 33%. Soluble sugars are both products of photosynthesis and osmolytes in their own right, so this result probably reflects the preserved chlorophyll feeding a steady carbon supply. Inoculation-driven accumulation of compatible solutes has been shown directly in rice (Arora et al., 2020; Ramasamy & Mahawar, 2023). Our data fit that picture, but they can't show whether the extra sugar was used for osmotic adjustment or simply reflects higher assimilation.

4.4 Consortia: complementarity is real but depends on the pair

The meta-analytic expectation is that consortia outperform single strains (Liu et al., 2023). Our results support a more guarded version of that claim. On the CMI averaged across salinity levels, only one consortium, C4-5, beat both of its parents. C2-5 sat between its parents, and C1-3, C1-5 and C2-3 ranked below both of theirs. Two consortia even fell below the uninoculated control on the average CMI. Mixing strains, then, isn't automatically an improvement.

C4-5 deserves a closer look, because it combined IS-04 and IS-05, the third- and second-ranked single isolates. At 150 mM, C4-5 exceeded both parents for soluble protein (181.51 vs. 133.45 and 138.26 $\mu\text{g mL}^{-1}$), chlorophyll *a* (15.6 vs. 12.5 and 11.6 mg g^{-1} FW) and SVI (2246 vs. 1766 and 2010). It fell below both for soluble sugars (70.08 vs. 78.77 and 82.25 $\mu\text{g mL}^{-1}$). A consortium that improves some traits beyond either parent, but not all, is what complementarity looks like in practice. Each partner probably contributes a different mechanism, and the combination covers more of the stress response than either strain alone. Consortium inoculation has likewise improved growth and yield of wheat under saline conditions (M. Y. Khan et al., 2022). Liu et al. (2023) linked the advantage of consortia partly to functional diversity among the members.

Why did the other pairs fail? The in vitro compatibility test was passed by all five, so the answer isn't simple antagonism on agar. Two explanations seem plausible. Strains that share a niche inside the root may compete for the same colonization sites, and the partner with weaker plant benefits may win (Compant et al., 2019; Kandasamy & Kathirvel, 2023). Alternatively, the same trait may be duplicated in both partners, adding cost to the plant without adding function. The pigment data offer a hint. Each of the three weak consortia contains at least one of the three isolates that protected chlorophyll least (IS-01, IS-02 and IS-03), two of them pair two such isolates, and all three lowered chlorophyll *a* below the

uninoculated control. The pattern isn't absolute: C2-5 pairs IS-02 with IS-05 and still protected chlorophyll *a*. Even so, if a poor partner can dilute a good one, mixing adds risk as well as potential.

We should add a qualification. The consortium and isolate experiments were run as separate batches with a shared uninoculated control, and the chlorophyll *b* and DPPH data show that batch effects existed. Comparisons between consortia and their parent isolates are therefore indirect. A single experiment testing each consortium next to its own two parents would give a cleaner test of synergy, and we see that as the logical next step.

4.5 Protein, phenolics and antioxidant activity

Protein retention was the one trait where consortia clearly outperformed single strains. C4-5 and C2-5 retained 81% and 75% of their unstressed protein at 150 mM, against 70% for IS-06 and 40% for uninoculated plants. Soluble protein reflects both synthesis and breakdown. Under salinity, poorer nitrogen assimilation and faster proteolysis both push it down (Parida & Das, 2005). A consortium with two partners could plausibly ease both processes, for example if one partner improves nitrogen supply while the other limits oxidative damage to proteins. Our data don't settle which process dominated. The strong protein–sugar correlation ($r = 0.89$) suggests that the two traits share a common driver, most likely the overall carbon and nitrogen status of the leaf.

Phenolics behaved like a stress-protective trait that the susceptible cultivar could not sustain on its own. Uninoculated leaves lost 70% of their phenolic content at 150 mM, while C4-5 kept its phenolic level near the unstressed control at 50 mM, and IS-06 did the same up to 100 mM. Phenolic compounds scavenge radicals and stabilize membranes, and bacterial inoculation is known to raise antioxidant capacity in salt-stressed plants (Neshat et al., 2022). The sharp drop under IS-01 at 150 mM ($2.80 \mu\text{g mL}^{-1}$) is an unexpected outlier. It suggests that some endophytes can suppress phenolic accumulation, and that such strains shouldn't be carried forward into formulations.

DPPH activity was the least informative trait. The unstressed control differed between batches (90% vs. 70.6%), activity in uninoculated plants was not monotonic with salt, and the trait correlated only weakly with the other markers. DPPH measures the combined radical-quenching capacity of an extract, not the activity of any one enzyme (Gülçin, 2020). It can therefore move for reasons unrelated to stress protection. Phenolic constituents and antioxidant activity of rice also respond to other abiotic stresses (Bhadwal et al., 2024), which

makes the assay a broad rather than a specific marker. The one pattern that did hold was that IS-06 and C4-5 gave the highest scavenging at 150 mM. We would treat DPPH as supporting evidence here and not as a primary endpoint. Measuring superoxide dismutase, catalase and ascorbate peroxidase activities directly would give a sharper picture of the antioxidant response.

4.6 What the indices and PCA add

Ranking inoculants trait by trait gives different winners depending on which trait is chosen: IS-06 for sugars, C4-5 for protein, IS-05 for root length. The CMI and the PCA resolve this by asking which treatments stay near the top across the whole profile. Both methods agreed on the first two places, IS-06 and C4-5, and on the bottom of the ranking. That agreement matters because the two methods weight traits differently. The CMI weights every trait equally, whereas PCA weights them by how much they vary together. Negrão et al. (2017) argue that composite, relative measures are the most defensible way to compare salt responses when baselines differ, and our experience supports that view.

PC1 captured half the variance and loaded positively on every trait. In other words, most of what distinguished good and poor treatments was a single, general axis of seedling health, rather than trade-offs between traits. PC2 then separated osmolyte-type traits (protein and sugars) from root and shoot length and DPPH activity. On this axis, the consortia and the best single isolates took different routes to the same outcome. The consortia leaned towards protein retention, while IS-06 leaned towards growth and pigment protection. Jolliffe and Cadima (2016) note that components with modest variance can still carry biologically meaningful structure, and PC2 seems to do so here.

4.7 Limitations and next steps

Several limitations need to be stated. First, the isolates were not identified to species, and their growth-promoting traits were not screened. Without 16S rRNA gene sequencing, the strains can't be compared with published inoculants or registered as bioinoculants. That work is essential before any field release (Afzal et al., 2019; Ganie et al., 2021). Second, biochemical values are expressed per millilitre of extract. The fixed tissue-to-solvent ratio makes treatments comparable, but it prevents direct comparison with studies that report per gram fresh weight. The absolute chlorophyll values should also be checked against a per-gram calculation before they are compared with other studies. Third, several assays recorded identical replicate SDs across all treatments (± 0.15). This probably reflects analytical rather

than biological replication, which would inflate the F values in Table 3; the effect sizes and ranking are more reliable than the exact *p*-values. Fourth, colonization was not verified. Re-isolating the inoculated strains from surface-sterilized roots and leaves, or tracking them with marker genes, would confirm that the effects came from endophytic populations (Walitang et al., 2017; Gupta et al., 2023). Finally, the trial ended at 45 DAS in pots. Salt tolerance at the seedling stage correlates only weakly with tolerance at flowering in rice (Singh et al., 2021), so yield-stage field trials remain the real test.

Despite these limits, the study offers a practical screening sequence: a compatibility check, a laboratory vigour assay, a graded-salinity pot trial and composite ranking. That sequence narrowed eleven candidate inoculants to two. For IS-06 and C4-5, we'd now prioritize strain identification, quantification of growth-promoting traits, a direct consortium-versus-parent comparison in one experiment, and multi-location field testing in salt-affected rice fields. Coupling inoculation with a tolerant cultivar is also worth testing, since host genotype shapes the endophytic community (Walitang et al., 2018; Wang et al., 2025).

5. Conclusions

Salinity at 150 mM NaCl reduced seedling vigour of IR-64 rice by 58%, chlorophyll *a* by 83% and soluble sugars by 68%. Bacterization with rice endophytes didn't change germination significantly, but it strongly protected seedling growth, photosynthetic pigments, soluble protein, sugars and phenolics under salt. Isolate IS-06 was the most consistent single strain. At 150 mM it raised chlorophyll *a* by 240%, soluble sugars by 169% and SVI by 142% over uninoculated plants. Consortium C4-5 retained 81% of its unstressed protein and exceeded both of its parent strains for protein, chlorophyll *a* and vigour. Three of the five consortia, however, ranked below both of their parent strains, so the benefit of combining strains depended on which strains were combined. A composite membership index and principal component analysis agreed on IS-06 and C4-5 as the leading candidates. The next steps are to identify these strains, confirm that they colonize the plant, and test them at the yield stage in saline fields.

657 **Declarations**

658 **Author contributions (CRediT).** [Author Name]: Conceptualization, Investigation, Formal
659 analysis, Data curation, Writing – original draft. [Co-author / Research Guide Name]:
660 Supervision, Methodology, Resources, Writing – review and editing. All authors read and
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667 **Data availability.** Treatment-level means, standard deviations and germination counts
668 underlying all tables and figures, together with the analysis scripts, are available from the
669 corresponding author on reasonable request and will be deposited in a public repository on
670 acceptance.

671 **Declaration of generative AI use.** During the preparation of this manuscript, the authors
672 used an AI writing assistant to help structure the text, run the statistical calculations from the
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